O zdarzeniach $A, B \subset \Omega$ wiadomo, że $P(A) = \frac{1}{4}, P(B) = \frac{1}{3}, P(A \cap B) = \frac{1}{5}$.
Oblicz $P(A' \cap B')$.

① DANE:

$$\begin{cases} P(A) = \frac{1}{4} \\ P(B) = \frac{1}{3} \\ P(A \cap B) = \frac{1}{5} \end{cases}$$

DO OBLICZENIA:

$P(A' \cap B') = ?$

② wiemy, że. $P(A' \cap B') = 1 - P(A \cup B)$

i $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

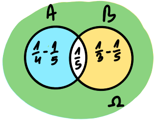
zatem: $P(A \cup B) = \frac{1}{4} + \frac{1}{3} - \frac{1}{5} = \frac{15}{60} + \frac{20}{60} - \frac{12}{60} = \frac{23}{60}$

więc: $P(A' \cap B') = 1 - \frac{23}{60} = \frac{37}{60}$

odp: $\frac{37}{60}$

II sposób rozwiązania przy pomocy diagramu Venna

$P(A' \cap B') = ?$



$P(A \cup B) = \frac{1}{4} + \frac{1}{3} - \frac{1}{5} = \frac{23}{60}$

$P(A' \cap B') = 1 - P(A \cup B) = 1 - \frac{23}{60} = \frac{37}{60}$